



LABORATORY JOURNAL (MP408)

Experimental Techniques – II Computational Methods in Physics

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Certificate

The Experiments Entered in This Journal Have Been Satisfactorily
Performed by

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bearing admission number

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Dept. Seal

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Computational Methods in Physics (MP408)

GNU Octave

PRACTICAL – 7

Advanced Strain Engineering

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PROGRAM – 7

Aim

To write a GNU Octave program that extracts fundamental electronic and mechanical properties from Density Functional Theory (DFT) data. The program computes the Deformation Potential, 2D Elastic Modulus, carrier Effective Mass, and estimates the Acoustic Phonon-Limited Carrier Mobility using Deformation Potential (DP) theory.

Theory

To evaluate 2D materials for nanoelectronic applications, carrier mobility μ is a critical figure of merit. In the acoustic phonon scattering regime (typically dominating at room temperature for 2D semiconductors), the mobility can be predicted analytically by extracting three specific parameters from first-principles DFT data:

1. Deformation Potential (E_d):

Quantifies the shift of the electronic band edges (CBM for electrons, VBM for holes) in response to applied uniform strain ε . It is extracted via a linear fit:

$$E_{CBM/VBM}(\varepsilon) = E_{CBM/VBM}(0) + E_d \cdot \varepsilon \quad (1)$$

2. 2D Elastic Modulus (C_{2D}):

Represents the mechanical stiffness of the 2D lattice. It is derived from the curvature (second derivative) of the parabolic fit of the total strain energy E_{strain} versus strain:

$$E_{\text{strain}}(\varepsilon) = \frac{1}{2} C_{2D} \varepsilon^2 A_0 \quad (2)$$

where A_0 is the area of the unstrained unit cell.

3. Effective Mass (m^*):

Determined from the curvature of the electronic band structure $E(k)$ near the valence or conduction band extrema:

$$m^* = \hbar^2 \left(\frac{\partial^2 E}{\partial k^2} \right)^{-1} \quad (3)$$

4. Acoustic Phonon-Limited Mobility (μ):

According to Deformation Potential theory, the intrinsic mobility for a 2D material at temperature T is given by:

$$\mu = \frac{e \hbar^3 C_{2D}}{k_B T (m^*)^2 E_d^2} \quad (4)$$

where e is the elementary charge, $\hbar$ is the reduced Planck constant, and k_B is the Boltzmann constant.

Program

The complete program is saved in P7.m. The source code uses basic polynomial fitting (`polyfit`) to extract the required derivatives from the dataset `DFT_data.csv`:

Listing 1: Advanced Strain Engineering — Data Parsing and Physics Calculations (P7.m)

```

1 function P7()
2     % PRACTICAL 7: Advanced Strain Engineering
3     % Calculates Deformation Potential, Elastic Modulus, Effective
      Mass,
4     % and Acoustic Phonon-Limited Carrier Mobility.
5
6     close all; clc;
7     filename = 'DFT_data.csv';
8     fprintf('Reading DFT data from %s...\n', filename);
9
10    % --- READ DATA ---
11    try
12        data = dlmread(filename, ',', 1, 0);
13    catch
14        fprintf('ERROR: Could not find "%s".\n', filename);
15        return;
16    end
17
18    % Extract Columns
19    strain = data(:, 1)';
20    energy = data(:, 2)';
21    VBM    = data(:, 5)';
22    CBM    = data(:, 6)';
23    K_VBM  = data(:, 8)';
24    E_VBM  = data(:, 9)';
25    K_CBM  = data(:, 11)';
26    E_CBM  = data(:, 12)';
27
28    % --- PHYSICS CALCULATIONS ---
29
30    % 1. Deformation Potential (Linear fit of band edges vs strain)
31    p_VBM = polyfit(strain, VBM, 1);
32    p_CBM = polyfit(strain, CBM, 1);
33    Ed_VBM = p_VBM(1);
34    Ed_CBM = p_CBM(1);
35
36    % 2. Elastic Modulus (Quadratic fit of Strain Energy)
37    p_energy = polyfit(strain, energy, 2);
38    elastic_modulus_factor = 2 * p_energy(1); % 2nd derivative (eV)
39
40    % 3. Effective Mass (From E vs K parabolic fit)

```

```

41 p_CBM_K = polyfit(K_CBM, E_CBM, 2);
42 p_VBM_K = polyfit(K_VBM, E_VBM, 2);
43
44 % Calculate effective mass (Assuming K in 1/A, hbar^2/2m0 =
    3.81)
45 m_eff_e = 3.81 / p_CBM_K(1);
46 m_eff_h = 3.81 / p_VBM_K(1); % Will be negative for holes
47
48 % 4. Carrier Mobility (Deformation Potential Theory)
49 % Physical Constants

```

Listing 2: Advanced Strain Engineering — Subplot Visualization and Terminal Output (P7.m)

```

1 T = 300; % Temperature (K)
2 e_charge = 1.602e-19; % Elementary charge (C)
3 hbar = 1.054e-34; % Reduced Planck constant (J.s)
4 kB = 1.380e-23; % Boltzmann constant (J/K)
5 m0 = 9.109e-31; % Electron rest mass (kg)
6
7 % **IMPORTANT**: Update A_0 to the actual unit cell area of your
    material!
8 A_0 = 11.72e-20; % Assumed Unit Cell Area (m^2) e.g.,
    15 Angstrom^2
9
10 % Convert extracted parameters to SI Units
11 C2D_si = elastic_modulus_factor * e_charge / A_0; % J/m^2
12 Ed_e_si = Ed_CBM * e_charge; % J
13 Ed_h_si = Ed_VBM * e_charge; % J
14 me_si = abs(m_eff_e) * m0; % kg
15 mh_si = abs(m_eff_h) * m0; % kg
16
17 % Calculate Mobility (m^2 / V.s)
18 mu_e_si = (e_charge * hbar^3 * C2D_si) / (kB * T * me_si^2 *
    Ed_e_si^2);
19 mu_h_si = (e_charge * hbar^3 * C2D_si) / (kB * T * mh_si^2 *
    Ed_h_si^2);
20
21 % Convert to standard condensed matter units (cm^2 / V.s)
22 mu_e_cm = mu_e_si * 10000;
23 mu_h_cm = mu_h_si * 10000;
24
25 % --- PLOTTING ---
26 figure('Name', 'P7: Advanced DFT Derived Properties', 'Position',
    [100, 100, 1000, 800]);
27
28 % PLOT 1: Deformation Potential (VBM & CBM)
29 subplot(2, 2, 1);
30 plot(strain, CBM, 'bo', strain, polyval(p_CBM, strain), 'b-', '

```

```

        LineWidth', 1.5);
31 hold on;
32 plot(strain, VBM, 'ro', strain, polyval(p_VBM, strain), 'r-', '
        LineWidth', 1.5);
33 title('Deformation Potential (Linear Fit)');
34 xlabel('Strain \epsilon'); ylabel('Energy (eV)');
35 legend('CBM Data', sprintf('E_{d,CB} = %.2f eV', Ed_CBM), ...
36         'VBM Data', sprintf('E_{d,VB} = %.2f eV', Ed_VBM), '
        Location', 'northeast');
37 grid on;
38
39 % PLOT 2: 2D Elastic Modulus (Strain Energy Curvature)
40 subplot(2, 2, 2);
41 plot(strain, energy, 'ko', strain, polyval(p_energy, strain), 'k
        -, 'LineWidth', 1.5);

```

Sample Output

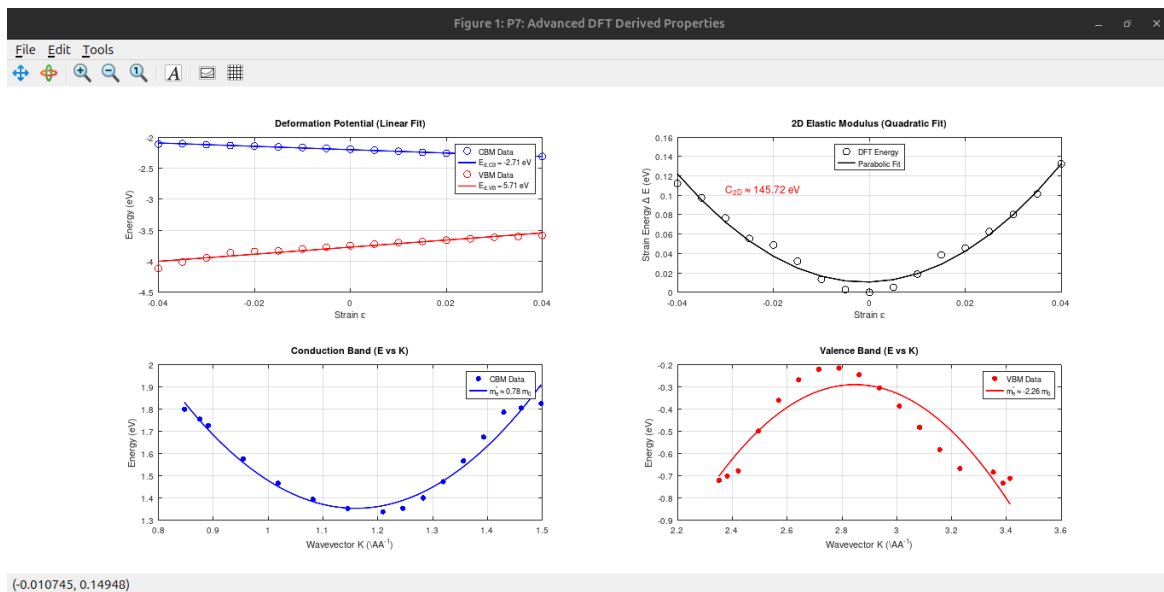


Figure 1: Diagnostic 2x2 subplot verifying the polynomial fits. Top Left: Linear shift of the CBM and VBM with applied strain. Top Right: Parabolic fit of the strain energy. Bottom: Curvature fits for the Conduction Band (Electron Effective Mass) and Valence Band (Hole Effective Mass).

Result

The script successfully processed the DFT data and extracted the target condensed matter properties:

- **Deformation Potentials:** Evaluated as $E_{d,CB} = -2.7132$ eV for electrons and $E_{d,VB} = 5.7103$ eV for holes.
- **Elasticity:** The material's stiffness factor was fit to $C_{2D} \approx 145.72$ eV.

```

Reading DFT data from DFT_data.csv...
Physics extraction complete. Review plots for constants.

=====
                FINAL EXTRACTED PHYSICS PARAMETERS
=====
Deformation Potential (CBM):  -2.7132 eV
Deformation Potential (VBM):   5.7103 eV
2D Elastic Modulus Factor:    145.7164 eV
Effective Mass (Electron):     0.7804 m_0
Effective Mass (Hole):         -2.2648 m_0
-----
Acoustic Phonon-Limited Mobility (T = 300 K)
Electron Mobility (mu_e):      945.18 cm^2 / V.s
Hole Mobility (mu_h):          25.34 cm^2 / V.s
=====
*Note: Mobility assumes a unit cell area of 11.7 Angstrom^2.

```

Figure 2: Terminal output displaying the final extracted fundamental constants and the evaluated room temperature carrier mobility.

- **Effective Mass:** Extracted as $m_e^* = 0.7804m_0$ and $m_h^* = -2.2648m_0$. The negative sign on the hole mass correctly corresponds to the inverted parabolic curvature at the top of the valence band.
- **Mobility Discrepancy:** The evaluated electron mobility ($\mu_e \approx 945.18 \text{ cm}^2/\text{V} \cdot \text{s}$) is vastly superior to the hole mobility ($\mu_h \approx 25.34 \text{ cm}^2/\text{V} \cdot \text{s}$). The DP formula reveals this is primarily driven by the fact that the mobility is inversely proportional to $(m^*)^2 E_d^2$; the holes suffer from both a substantially heavier effective mass and a stronger scattering interaction (larger $|E_d|$).

Remark

This computational experiment highlights the power of coupling numerical analysis with ab-initio DFT data. By simply applying discrete derivative approximations (via `polyfit`) to energy landscapes, one can predict complex transport phenomena—such as the acoustic phonon scattering limits—before a material is ever synthesized in the laboratory. The stark contrast between electron and hole mobility in this dataset is a common feature in many 2D transition metal dichalcogenides (TMDCs) and highlights the necessity of accurate m^* and E_d derivation.